STORAGE-BENCH · SUPABASE STORAGE · SMOKE RUN

Supabase Storage on storage-bench

Supabase Storage, reached through its S3-compatible endpoint, run through the same harness as the published storage-bench results (files-sdk 3.0.0, Node 24, no retries), from four Fly.io machines placed as close as Fly allows to the original client locations.

> **This is a smoke run, not a benchmark result.** Each Supabase number comes from 1 to 3 samples; the published numbers come from 26 to 180. The Supabase bucket is in Frankfurt while every other provider's bucket is in US East, so the ranking follows geography more than service speed. Page 3 explains how the two setups differ.

228 / 228
requests succeeded across the four regions, with no retries

13 of 13
timed operations exercised: upload, head, download, delete, presigned GET and PUT, range read, copy, listing, parallel and burst uploads, fresh-client HEAD

4 regions
Fly iad, sjc, ams and cdg, matching Ashburn, San Francisco, Amsterdam and Lille

Amsterdam at a glance

Amsterdam is the one location where both setups ran from the same city, and our closest client to the Frankfurt bucket (365 km, against about 6,200 km to the US East buckets). Supabase median from Fly ams, next to the fastest of the five published providers from Railway's Amsterdam server.

Read a small file	Upload a photo	Download a photo	First request	Gallery upload
111 ms	**115 ms**	**124 ms**	**79 ms**	**389 ms**
Fastest other provider: Cloudflare R2, 200 ms	Fastest other provider: Cloudflare R2, 461 ms	Fastest other provider: Cloudflare R2, 264 ms	Fastest other provider: Cloudflare R2, 146 ms	Fastest other provider: Railway, 378 ms **(faster than Supabase)**

Medians. Supabase: Fly ams, 16 Oct 2026, smoke profile. Others: published storage-bench results from railway-amsterdam, full profile, three passes on 9 and 15 Oct 2026.

What this shows

- **The S3 path works end to end.** Every timed operation the harness runs succeeded against Supabase from all four regions, with no retries.
- **In Europe, Supabase is ahead because its bucket is nearby.** From Amsterdam and Paris it leads every test except 16 parallel uploads, where Railway is slightly faster from both (by 12 ms and 15 ms). The other buckets are about 6,200 km away.
- **In the US, it trails for the same reason.** From Ashburn, Supabase is the slowest on every test; the other buckets are within about 485 km, while Frankfurt is 6,500 km away.
- **A fair comparison needs a us-east-1 project** and full runs, as set out on the last page.

AMSTERDAM IN DETAIL

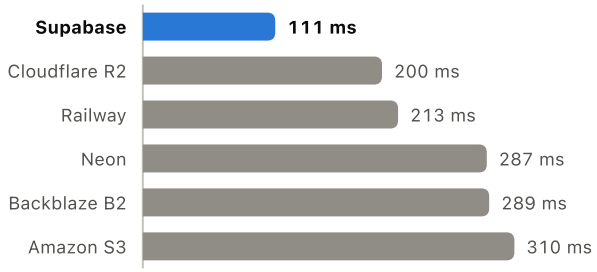
Five everyday operations from Amsterdam

Each chart ranks the six providers for one operation. Supabase's bucket is about 365 km away; the others are across the Atlantic. Even so, Railway edges Supabase on 16 parallel uploads (378 ms against 389 ms, from a single Supabase sample).

Supabase (bucket in Frankfurt, eu-central-1) Published providers (buckets in US East)

Read a small file

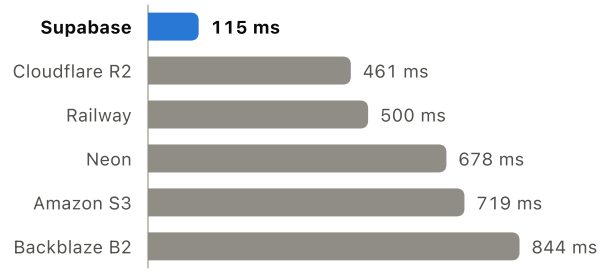
1 KB presigned GET, time to first byte. Median, lower is better.



Samples: Supabase 3, other providers 178–180 each.

Upload a photo

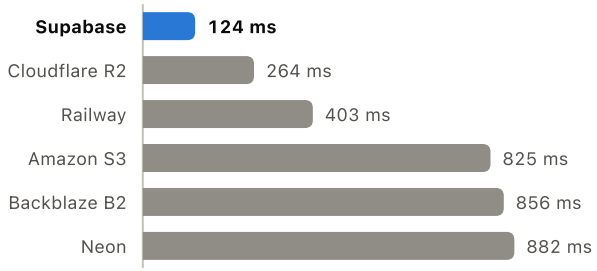
1 MB upload with put(). Median, lower is better.



Samples: Supabase 2, other providers 90 each.

Download a photo

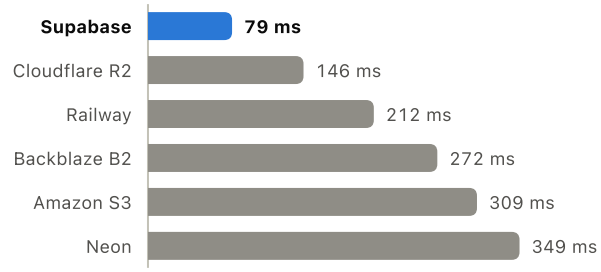
1 MB presigned GET, whole body. Median, lower is better.



Samples: Supabase 2, other providers 90 each.

First request

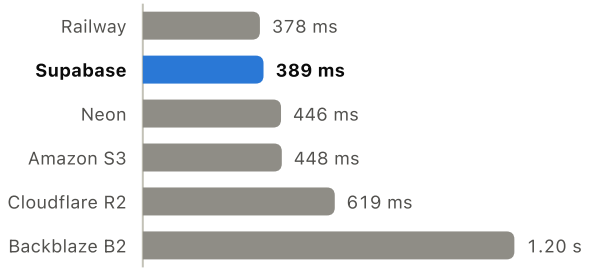
Fresh client: new TCP and TLS, one HEAD. Median, lower is better.



Samples: Supabase 2, other providers 60 each.

Gallery upload

16 uploads of 100 KB in parallel. Median, lower is better.



Samples: Supabase 1, other providers 28–30 each.

Why Amsterdam

Both setups have a client in Amsterdam (Railway europe-west4 upstream, Fly ams here), so this is the most direct comparison available. By distance it is also our closest client to Frankfurt.

Paris (Fly cdg) is about 115 km farther from Frankfurt, yet it measured lower latency in this run: 54 ms against 111 ms to the first byte of a 1 KB read. Network routes, not straight-line distance, decide.

SETUP

How our setup differs from the original

Same harness, same SDK, same operations and the same no-retry rule. What differs is where the client machines run, which networks they use, where the bucket is, and how many samples were taken.

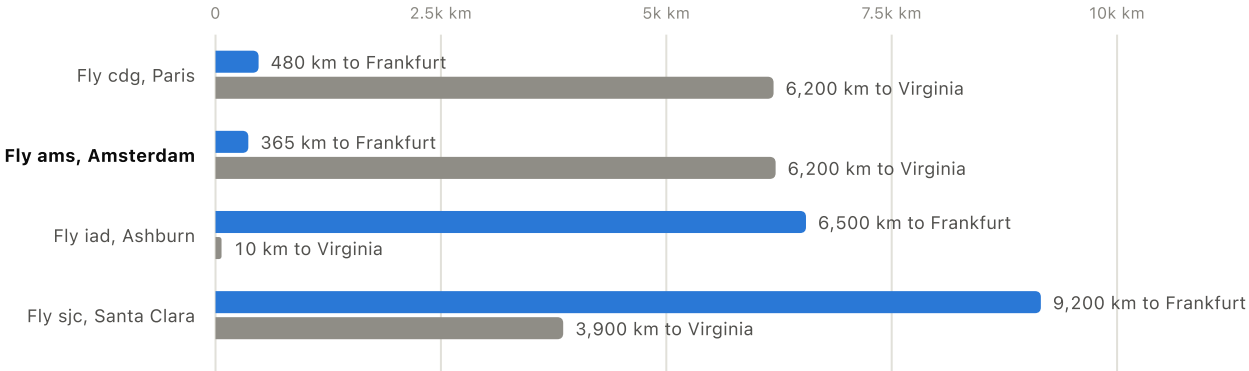
Client locations

Original client	Host and network	Our client	Host and network	Apart
Ashburn, Virginia	Railway (us-east4), AS400940 Railway	Fly iad, Ashburn	Fly.io; ipinfo: AS60068 Datacamp Limited	same metro
San Francisco	Freestyle VM, AS36320 Freestyle Cloud	Fly sjc, Santa Clara	Fly.io; ipinfo: AS396356 Latitude.sh	60 km
Amsterdam	Railway (europe-west4), AS400940 Railway	Fly ams, Amsterdam	Fly.io; ipinfo: AS60068 Datacamp Limited	same metro
Lille, France	boat sandbox on OVH, AS16276 OVH	Fly cdg, Paris	Fly.io; ipinfo: AS30081 CacheFly	205 km
Seattle	Tenki VM, Wowrack then Luxor	none	Fly has no Seattle region	–
Florida home line	Home server, AS7018 AT&T	none	A residential line; no data center stands in for it	–

Bucket locations

Provider	Bucket region	From Amsterdam	From Ashburn
Amazon S3	us-east-2, Ohio	6,500 km	485 km
Cloudflare R2	Eastern North America location hint; R2 picks the site	n/a	n/a
Railway Buckets	iad, Virginia	6,200 km	same metro
Backblaze B2	us-east-005, Virginia	6,200 km	same metro
Neon	aws us-east-1, Virginia	6,200 km	same metro
Supabase Storage	aws eu-central-1, Frankfurt	365 km	6,500 km

Distance from each of our clients to the buckets



Great-circle distances between city centers. Latency follows network routes, which are longer.

ALL REGIONS

Every location, every provider

Medians, lower is better. The fastest value in each row is underlined. Amsterdam is highlighted.
*Supabase: smoke run from the Fly region listed on page 3, 1 to 3 samples per value.

Read a small file 1 KB presigned GET, time to first byte

From	S3	R2	Railway	B2	Neon	Supabase*
Ashburn	42 ms	76 ms	**8 ms**	11 ms	41 ms	**203 ms**
San Francisco	**110 ms**	193 ms	123 ms	221 ms	146 ms	**222 ms**
Amsterdam	310 ms	200 ms	213 ms	289 ms	287 ms	**111 ms**
Lille	309 ms	198 ms	198 ms	283 ms	372 ms	**54 ms**

Upload a photo 1 MB upload with put()

From	S3	R2	Railway	B2	Neon	Supabase*
Ashburn	170 ms	257 ms	113 ms	**52 ms**	115 ms	**421 ms**
San Francisco	531 ms	**444 ms**	509 ms	618 ms	637 ms	**533 ms**
Amsterdam	719 ms	461 ms	500 ms	844 ms	678 ms	**115 ms**
Lille	1.06 s	715 ms	555 ms	895 ms	1.10 s	**185 ms**

Download a photo 1 MB presigned GET, whole body

From	S3	R2	Railway	B2	Neon	Supabase*
Ashburn	48 ms	137 ms	52 ms	**29 ms**	49 ms	**287 ms**
San Francisco	553 ms	**265 ms**	309 ms	609 ms	796 ms	**633 ms**
Amsterdam	825 ms	264 ms	403 ms	856 ms	882 ms	**124 ms**
Lille	879 ms	334 ms	436 ms	819 ms	954 ms	**85 ms**

First request Fresh client: new TCP and TLS, one HEAD

From	S3	R2	Railway	B2	Neon	Supabase*
Ashburn	79 ms	77 ms	20 ms	**13 ms**	21 ms	**137 ms**
San Francisco	228 ms	167 ms	**108 ms**	217 ms	321 ms	**216 ms**
Amsterdam	309 ms	146 ms	212 ms	272 ms	349 ms	**79 ms**
Lille	309 ms	166 ms	219 ms	284 ms	347 ms	**76 ms**

Gallery upload 16 uploads of 100 KB in parallel

From	S3	R2	Railway	B2	Neon	Supabase*
Ashburn	**114 ms**	352 ms	174 ms	888 ms	133 ms	**1.11 s**
San Francisco	**348 ms**	590 ms	445 ms	1.14 s	444 ms	**863 ms**
Amsterdam	448 ms	619 ms	**378 ms**	1.20 s	446 ms	**389 ms**
Lille	436 ms	796 ms	**391 ms**	1.13 s	443 ms	**407 ms**

OTHER DIFFERENCES AND NEXT STEPS

Everything else that differs

Beyond the regions on page 3.

- **Same:** files-sdk 3.0.0 with the AWS SDK S3 client, identical operations and object sizes, fresh random bodies with SHA-256 checks, one attempt per request.
- **Runtime:** Node 24.22.0 on Fly; the original runs used Node 24.14 to 24.21.
- **Machines:** Fly performance-1x (one dedicated CPU, 2 GB), against Railway servers, VMs, a sandbox and a home server.
- **Samples:** smoke profile (1 to 3 per number, no 100 MB tests, 100-object listing) against the full profile (10 to 60 per pass, three passes per location).
- **Timing:** Supabase ran alone on 16 Oct; the other providers ran together on 9 and 15 Oct, interleaved in a random order every round.
- **Untested:** 100 MB multipart uploads. The project's upload size limit has to be raised to at least 100 MB first.

Getting to a fair comparison

The harness works against Supabase end to end. Before the numbers can stand next to the published results, four things need to change.

- **Put the bucket in us-east-1.** Every other provider stores data in US East (Ohio or Virginia). A Supabase project in aws us-east-1 sits in the same region as Neon and in the same metro as Railway and B2.
- **Run the full profile.** Three full passes per location, as in the original runs, so every median rests on 26 to 180 samples, like the published ones, and the 95th and 99th percentiles are reported.
- **Run a reference provider on the same machines.** Our clients use different hosts and networks than the original ones. Running at least one of the published providers (for example an S3 bucket in us-east-2) alongside Supabase, in the same process and the same random order, separates provider differences from client differences.
- **Raise the upload size limit to 100 MB or more**, so the 100 MB multipart upload and download tests run instead of failing.

What cannot be matched on Fly

Fly has no Seattle region, and no data center reproduces a residential connection like the original Florida home line. Those two locations need other hosts, or are left out of the comparison.

Sources

Published numbers: results/summary.json in storage-bench (commit f38ad93). Supabase numbers: raw samples in results/raw-supabase-euc1, all four runs started at 07:58 UTC on 16 Oct 2026. Client network names come from ipinfo.io as recorded by the harness.